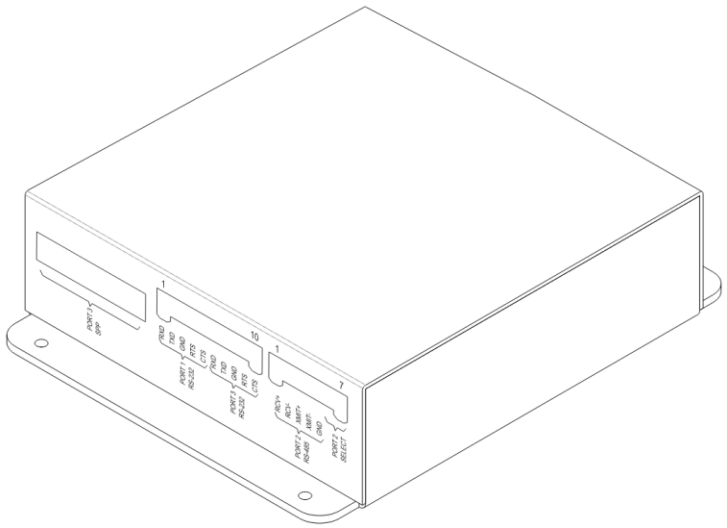
SIEMENS



VESDA Peripheral Module

Installation Instructions Model VPM

Installation

Mounting

A6V10347523_en--_g

2025-03-05

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1 Introduction

The Model VPM from Siemens Industry, Inc. is an HNET module that interfaces to a VESDA® HLI High Level Interface. The VPM must be mounted on a VPM-MP mounting plate (S54430-F95-A2) inside a CAB enclosure with an HLI card and a socket card (VESDA-HLI-Kit S54430-F99-A2). See Table 1 for a list of compatible VESDA® detectors and the limitations on the number of detectors per HLI. One VPM can support two VESDA-HLI-Kits and a total of 200 VESDA® detectors.

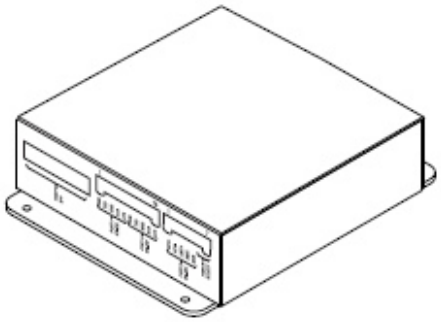


Fig. 1: VPM VESDA Interface Module

The VPM uses one network address on the FireFinder-XLS/Desigo Fire Safety Modular/Cerberus PRO Modular System. All VESDA® detectors connected to the VPM are addressed using a VESDA® programmer. For ease of use and troubleshooting purposes, be sure that the address assigned to the VESDA® detector in Zeus agrees with the address assigned to it by the VESDA® programming tool. The VPM communicates the status of the VESDA® devices over HNET to the FireFinder-XLS/Desigo Fire Safety Modular/Cerberus PRO Modular System where it can be displayed on the Operator Interface (OI).

1.1 Features

VPM features are as follows:

- Can be connected to the HNET either Style 4 or Style 7.
- Includes diagnostic LEDs to indicate failure of the HNET or the CPU. It also has a power on indicator.
- Includes a reset switch in the event that the VPM requires a hardware reset.
- Mounts inside any CAB enclosure on a VPM-MP mounting plate.
- Supports one or two VESDA® HLIs.
- VPM software can be upgraded in the field from a PC that has the Zeus program installed.

2 Operation

The VPM initializes, operates, and maintains all devices residing on the VESDAnet® network. The VPM communicates all relevant device and event information, such as alarms and troubles, to the OI. The sensitivity of any VESDA® detector can be checked from the OI through the VPM. All information about the devices on the loop can be displayed on the OI.

The VPM supports two HLI interface cards. Up to 100 VESDA® detectors can connect to each HLI. The VESDAnet® microprocessor controls the on-board isolator to isolate either zone from the VESDAnet® loop if one of them is shorted. When one zone is isolated from the loop, the other zone will still work.

2.1 Controls and Indicators

The HNET/UFP side panel of the VPM contains one reset switch, three LEDs, one termination switch and one HNET address switch as shown in Figure 2.

A reset switch is located on the top of the panel. Pushing the reset switch re-initializes the VPM operation.

POWER	(Green)	Normally ON. When illuminated, indicates that power for the VPM is applied to the module.
CPU FAIL	(Yellow)	Normally OFF. When illuminated, indicates that the module microprocessor has failed.
HNET/UFP FAIL	(Yellow)	Normally OFF. When illuminated, indicates that the HNET/UFP communication with the VPM has terminated.

A three-position switch located directly beneath the LEDs on the HNET side of the VPM is used to set the HNET network address of the VPM.

3 Pre-Installation

Before connecting either the HLI, power or the HNET, the network address must be set for the VPM using the three-position switch. (Refer to Figure 2 for the location of the switch.) The address for the VPM must be the same as the address selected for it in the Zeus Programming Tool. To increment each digit of the address, press the “+” button above the desired digit; to decrement each digit, press the “-” button below the desired digit. The range of allowable addresses is from 001 to 251 (leading zeros must be used).

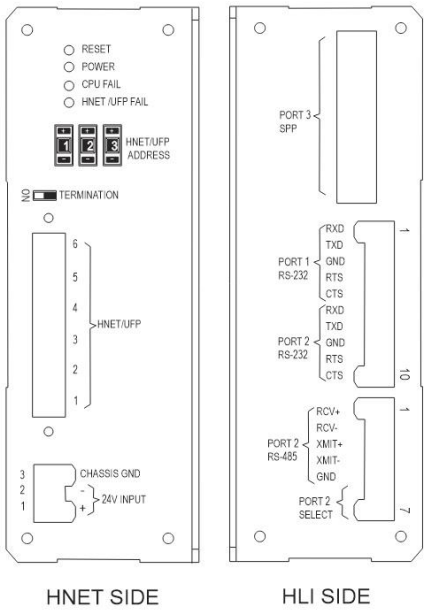


Fig. 2: VPM Side Panel Detail

NOTE: If the VPM is located at the end of the HNET network (Style 4 only), the termination switch must be set to ON. Otherwise, it must be set to OFF.

NOTE: To enable Port 2 (RS-232), the Port 2 Select pins (6 and 7) must be jumpered on the Port 2 RS-485 connector.

:

3.1 Configuring VESDAnet for use with the VPM

In order to comply with UL/ULC timing requirements for supervision of the VESDA detectors on VESDAnet, two supervision parameters must be set. While logged on and configuring the VESDAnet system with the Xtralis VSC program, perform the following steps:

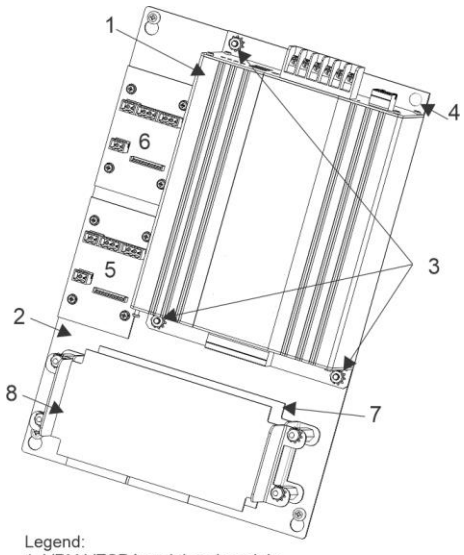
1. In the tree view, right click on “VESDAnet” (the root of the tree).
2. Select “edit” from the pop-up menu.
3. In the pop-up window, select the “Network Settings” tab.
4. Set the “Health check period (sec)” to 45.
5. Set the “Network delay period (sec)” to 15.
6. Click OK.
7. A warning pop-up will appear. Click OK.

It is also recommended to set the system date and time while configuring the VESDAnet.

1. In the tree view, right click on “VESDAnet.”
2. Select “device” from the options list.
3. Select “Select system date and time.”
4. In the pop-up window, set the current date and time.
5. Click OK.

4 Mounting

Mount the VPM to the VPM-MP on the upper right side of the bracket, as shown in Figure 3. One VPM-MP uses one CC-5 slot. Refer to the VPM-MP installation instructions, P/N A6V10347525, for information on mounting the VESDA-HLI-KIT (HLI high level interface card, socket card and cables) on the VPM-MP.



Legend:
1 VPM VESDA peripheral module
2 VPM-MP mounting plate
3 3 #8-32 kep lock hex nuts
4 Open mounting hole
5 Socket card #1
6 Socket card #2
7 HLI high level interface card #1
8 HLI high level interface card #2

Fig. 3: Mounting the VPM on the VPM-MP

1. Place the VPM on the three studs provided for it on the VPM-MP.
2. Place the three #8-32 kep lock hex nuts provided with the VPM-MP over the studs and secure the VPM in place on the mounting plate.
3. Leave the upper right hand mounting hole open. This hole will be used when the VPM-MP is installed into the CAB enclosure.

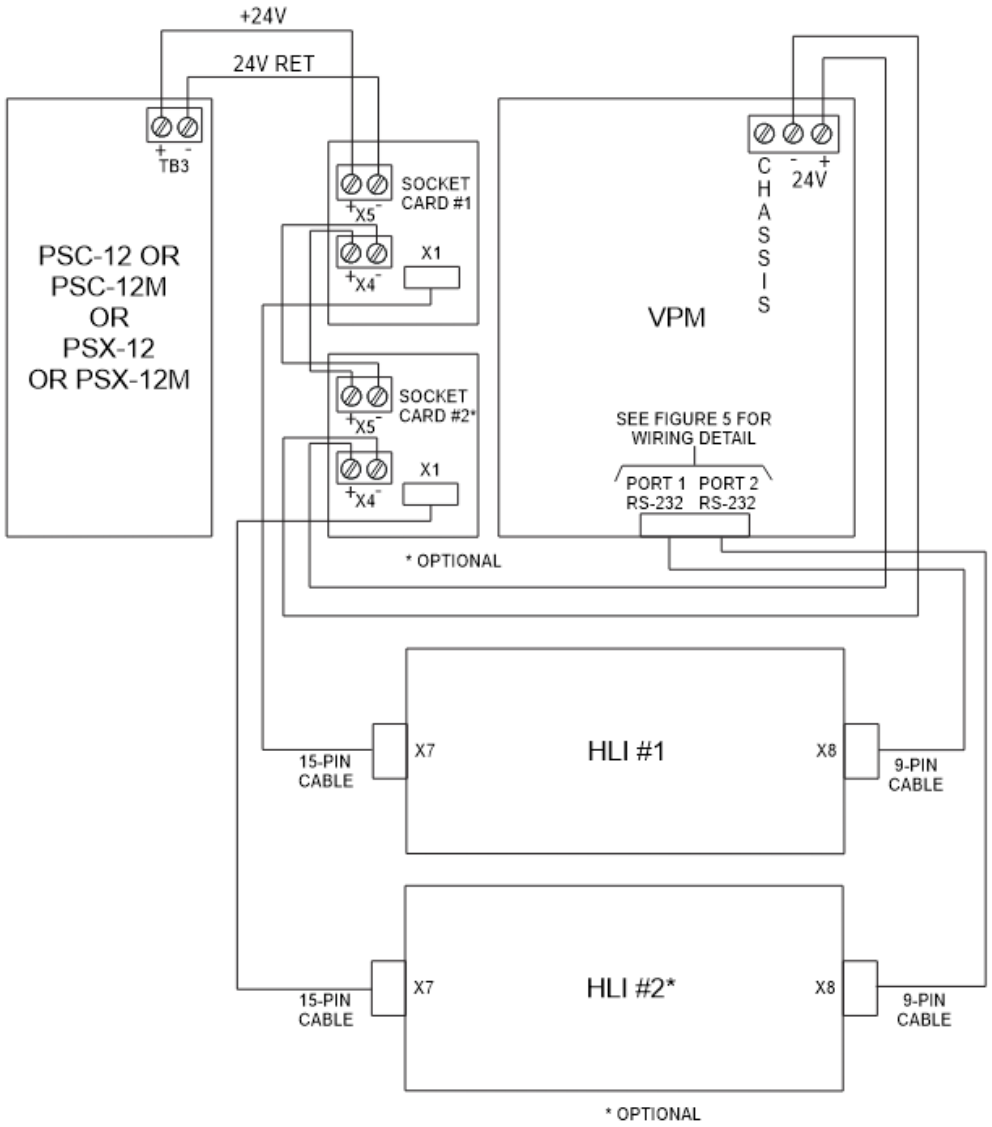


Fig. 4: Powering The VPM / HLI(s) From A PSC-12 or PSC-12M/PSX-12 or PSX-12M

NOTES:

1. 18 AWG min., 12 AWG max.
2. Power limited to NFPA72 per NEC 760.
3. No end of line device required.
4. . Reference the PSC-12 Installation Instructions, P/N 315-033060, the PSC-12M Installation Instructions, Document ID A6V12021370, the PSX-12 Installation Instructions, P/N 315-031120, or the PSX-12M Installation Instructions, Document ID A6V12021371.

NOTE: Only one power supply. PSC-12 or PSC-12M or PSX-12 or PSX-12M is allowed

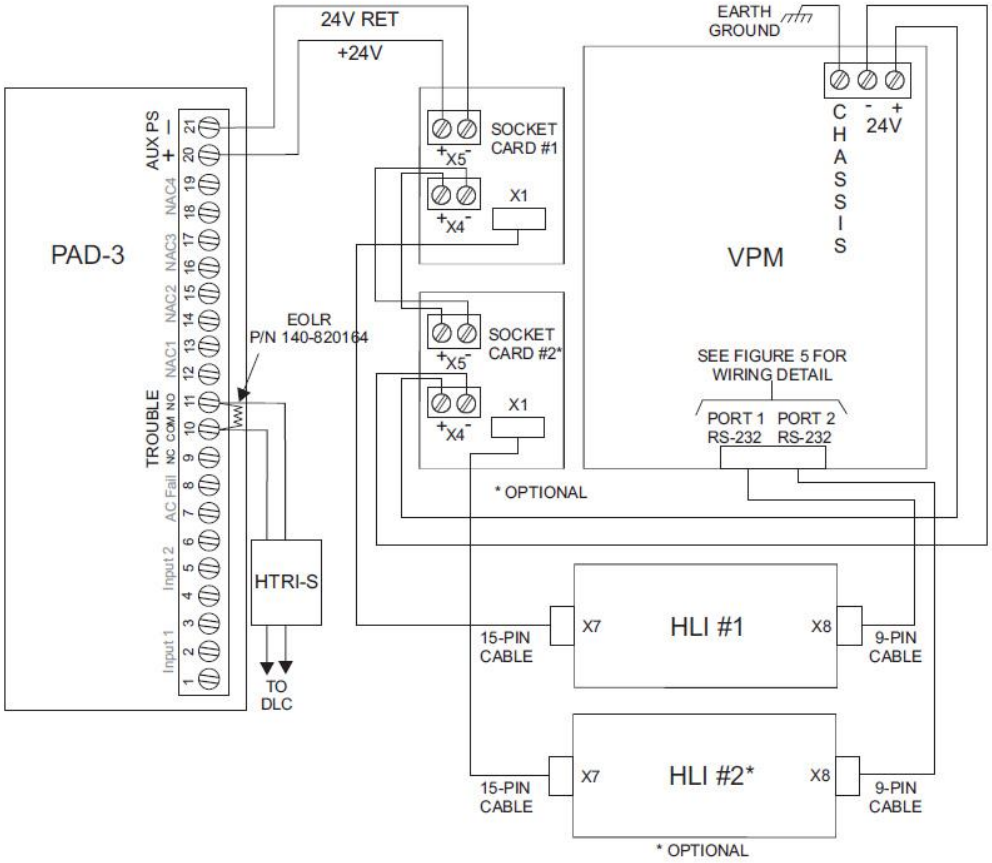


Fig. 5: Powering The VPM / HLI(s) From A PAD-3

NOTES:

1. 18 AWG min., 12 AWG max.
2. Power limited to NFPA72 per NEC 760.
3. No end of line device required.
4. Reference PAD-3 Installation Instructions, P/N 315-099082.
5. 5. Reference HTRI-S Installation Instructions, P/N 315-049475.
6. In Zeus, set the HTRI device usage to trouble.

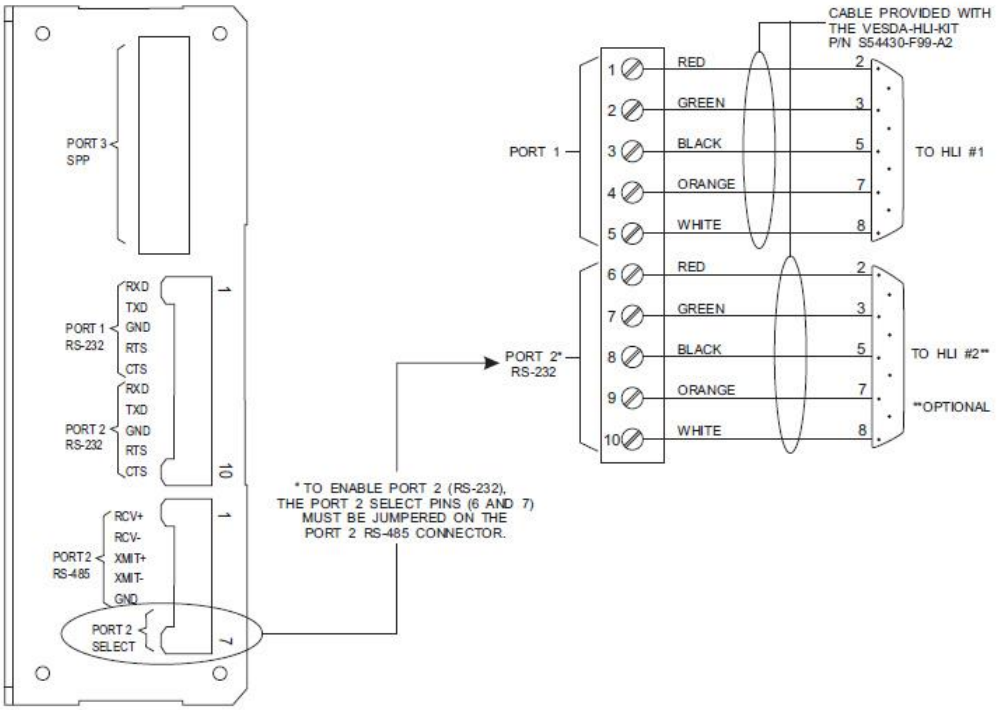


Fig. 6: VPM to HLI Connection

NOTES:

1. 4000 feet maximum between devices on the VESDAnet® network.
2. Total VESDAnet® length must not exceed 16,000 feet.
3. Recommended cable: Belden 9841 or equivalent.
4. Supervised, Style 4 (Class B) (DCLB).
5. Power limited to NFPA72 per NEC 760.
6. Ground Fault detected at 5 ohms or less.

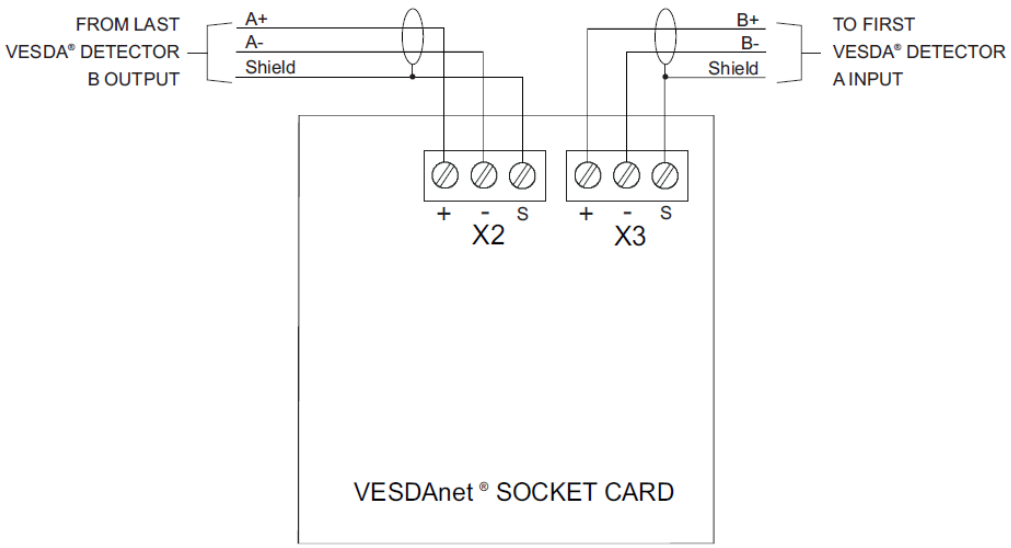


Fig. 7: Connection to VESDAnet Network

4.1 VESDAnet Limitations

The number of VESDA® detectors allowed on a single HLI is limited by two factors, the number of VESDAnet® detector nodes (100 maximum) and the number of VESDA® elements (252 maximum per VPM). Table 1 lists the number of VESDA® nodes and elements for each VESDA® detector model.

To determine if your configuration falls within the limits of an HLI, first add up the number of VESDA® detector nodes (see Table 1) that you plan for this HLI. This value must be <= 100.

Next, for each VESDA® detector planned for this HLI locate the number of elements it occupies in Table 1 and sum all these elements. This value must be <= 252 per VPM.

If either of these values are exceeded, you must remove detectors from the HLI until both values are within the limits. Add additional HLIs until each HLI in the system satisfies these requirements.

Device	Model No.	Nodes	Elements	Device	Model No.	Nodes	Elements
LaserFOCUS	VLF-250*/**	1	1	**LaserSCANNER Detectors**	VLS-200	1	5
	VLF-500*	1	1		VLS-300	1	5
LaserCOMPACT	VLC-505	1	1		VLS-600	1	5
LaserPLUS	VLP-400	1	1		VLS-700	1	5
	VLP-010	1	1		VLS-204	1	5
	VLP-002	1	1		VLS-304	1	5
	VLP-012	1	1		VLS-310	1	5
VESDA-E	VEP-A00-1P****+	1	1		VLS-214 1	1	5
	VEP-A00-P***+	1	1		VLS-314	1	5
	VEP-A10-P***+	1	1		VLS-210	1	5
	VEU-A00***+	1	1	**E-Series SCANNER**	VES-A00-P*****	1	5
	VEU-A10***+	1	1		VES-A10-P*****	1	5
VESDA-E (7th edition)	VEP-A00-1P-UL	1	1		VES-A00-P-UL	1	5
	VEP-A00-P-UL	1	1		VES-A10-P-UL	1	5
	VEP-A10-P-UL	1	1				
	VEU-A00-UL	1	1				
	VEU-A10-UL	1	1				
LaserINDUSTRIAL	VLI-885**	1	1				
FireTRACER	VFT-15**	16	16				

Table 1: VESDAnet® COMPATIBLE DEVICES

*Requires VIC-010 VESDAnet® Card.

**On these detectors, a jumper must be installed between the VESDAnet® A pair and B pair shield.

***The models VEP-A00-P, VEP-A10-P, VEU-A00, and VEU-A10 are recognized by XLS/Desigo Fire Safety Modular/Cerberus PRO Modular fire panels as models VLP. See Table 2 for details. VPM v1.01 supports these models in mimic mode and non-mimic mode. See Zeus user manual for more details.

****The model VEP-A00-1P is recognized by XLS/Desigo Fire Safety Modular/Cerberus PRO Modular fire panels as models VLC. See Table 2 for details. VPM v1.01 supports this model in mimic mode and non-mimic mode. See Zeus user manual for more details. *****The models VES-A00-P and VES-A10-P are recognized by XLS/Desigo Fire Safety Modular/Cerberus PRO Modular fire panels as models

+The models VEP-A00-1P, VEP-A00-P, VEP-A10-P, VEU-A00, and VEU-A10 are recognized by XLS/Desigo Fire Safety Modular/Cerberus PRO Modular fire panels using its own model number when operating with the user interface PMI/PMI-2/PMI-3 (XLS)/FCM2041-U2 (Desigo Fire Safety Modular)/FCM2041-U3 (Cerberus Pro Modular) and the Zeus Programming Tool version 15.01.xx.xx or higher.

NOTE: It is recommended to configure the VFT-15 in a non-latching mode. If latching is desired, the detectors must be manually reset prior to exiting walk test

VESDA E model	Siemens P/N (VESDA E)	VESDA E Description	Displayed in Zeus as	VLP equivalent model	VLP Siemens P/N	VLP Description
VEP-A00-1P	S54333-Z30-A1	VEP with LEDs, 1 Pipe	VEP-A00-1P *VLC w/ Zone #	VLC-505	500-699893	VESDA VLC-505 LASER Compact 1-Pipe Networkable via VESDA Net
VEP-A00-P	S54333-Z28-A1	VEP with LEDs	VEP-A00-P *VLP w/ Zone #	VLP-400	500-699818	VESDA VLP-400 LaserPLUS Detector with Fire & OK status LED lights
VEP-A10-P	S54333-Z29-A1	VEP with 3.5" Display	VEP-A10-P *VLP w/ Zone #	VLP-002	500-699819	VLP-002 LaserPLUS DETECTOR (includes detector and integral Display Module
VEU-A00	S54333-F1-A1	VEU with LEDs	VEU-A00 *VLP w/ Zone #	VLP-400	500-699818	VESDA VLP-400 LaserPLUS Detector with Fire and OK status LED lights
VEU-A10	S54333-F32-A1	VEU with 3.5" Display	VEU-A10 *VLP w/ Zone #	VLP-002	500-699819	VLP-002 LaserPLUS DETECTOR (includes detector and integral Display Module)
VES-A00-P	S54333-Z34-A1	VESDA-E Scanner with LED display and relays	*VLS w/ Zone #	VLS-700	500-699824	SCAN HEAD/12-REL FOK LED display
VES-A10-P	S54333-Z35-A1	VESDA-E Scanner with 3.5" LCD Display	*VLS w/ Zone #	VLS-304	500-699826	SCAN HEAD/DIS/12 RLY - Full Display

Table 2: Zeus Programming Equivalents

*Displayed when system is configured in mimic mode.

4.2 HNET Connection

The VPM can be connected to the HNET either Style 4 or Style 7. Refer to the Zeus configuration for the proper Style. In either case, the VPM can be connected in the middle of an HNET. For a Style 4 network, the VPM can also be connected at the end. If the VPM is connected at the end, care must be taken to properly terminate the HNET See Figures 8 and 9 for the wiring instructions when the VPM is in the middle and Figure 10 when the VPM is at the end of the network.

NOTE: In a single enclosure system the VPM requires an NIC-C in this enclosure.

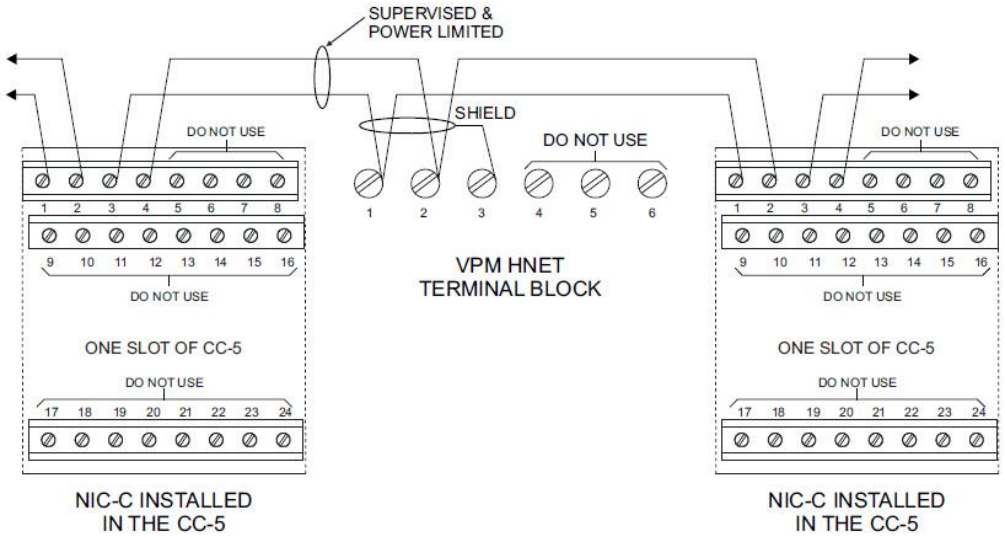


Fig. 8: VPM In The Middle Of A Style 4 HNET

NOTES:

1. No end of line resistor required. VPM termination set to OFF.
2. 18 AWG min., 12 AWG max.
3. 2000 feet max. between NIC-Cs.
4. Use twisted pair or shielded twisted pair.
5. Terminate shields at one end only.
6. Power limited to NFPA72 per NEC 760.
7. Reference NIC-C Installation Instructions, P/N 315-033240 for wiring and ground fault detection impedance.

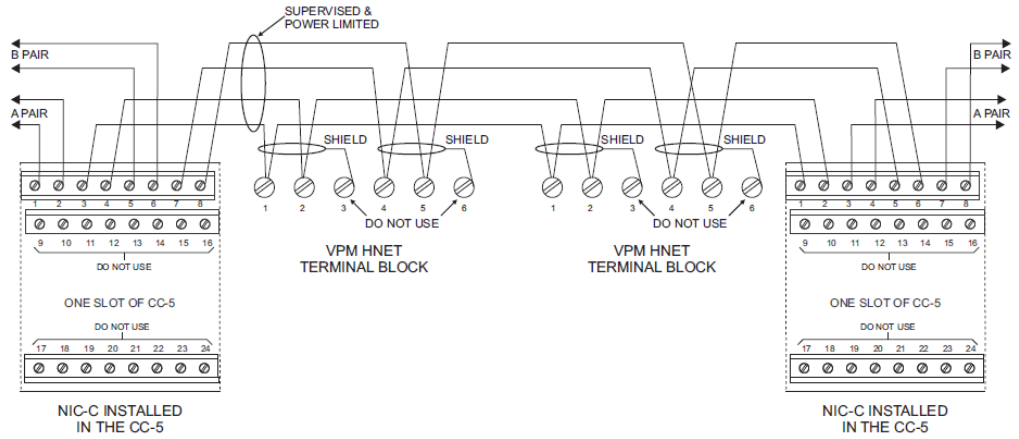


Fig. 9: VPM In The Middle Of A Style 7 HNET

NOTES:

1. No end of line resistor required. VPM termination set to OFF.
2. 18 AWG min., 12 AWG max.
3. 2000 feet max. between NIC-Cs.
4. Use twisted pair only.
5. Terminate shields at one end only.
6. Power limited to NFPA72 per NEC 760.
7. Reference NIC-C Installation Instructions, P/N 315-033240.

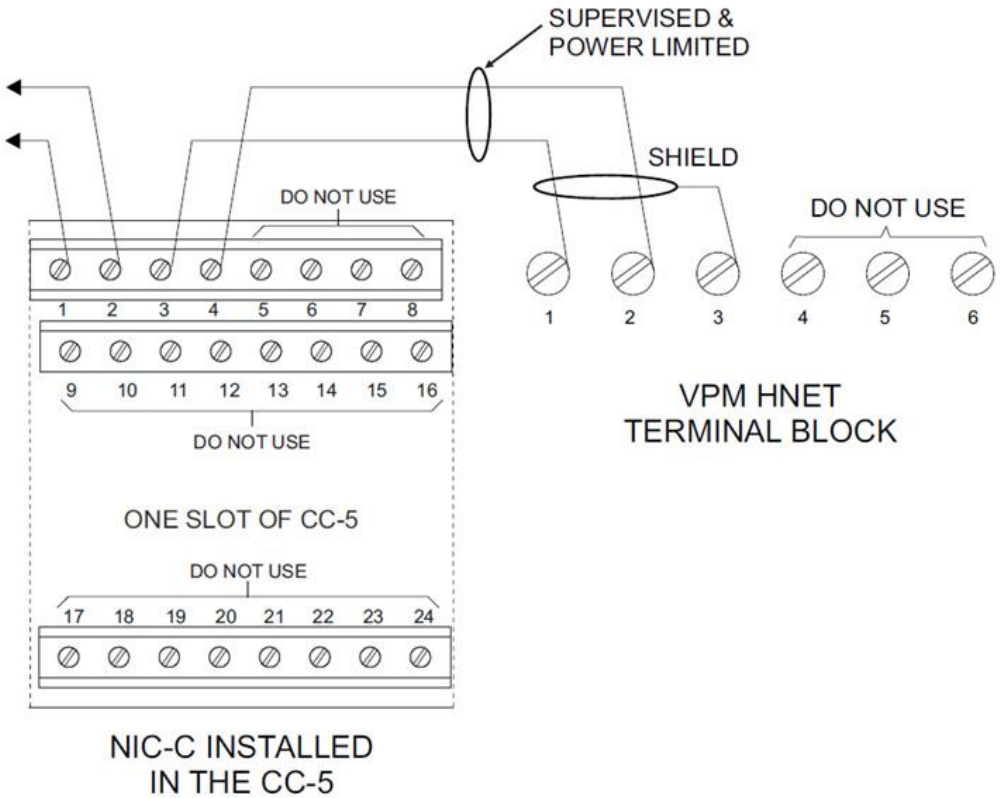


Fig. 10: VPM At The End Of A Style 4 HNET

NOTES:

1. No end of line resistor required. VMP termination set to ON.
2. 18 AWG min., 12 AWG max.
3. 2000 feet max. for NIC-C.
4. Use twisted pair or shielded twisted pair.
5. Terminate shield at one end only.
6. Power limited to NFPA72 per NEC 760.
7. Reference NIC-C Installation Instructions, P/N 315- 033240 for wiring and ground fault detection impedance.

5 Installation

The VPM mounts on the VPM-MP along with up to two HLI modules and 2 socket cards. The VPM-MP mounts in any CAB (CAB1, CAB1-X, CAB2 or CAB3) enclosure where a CC-5 can be mounted. The VPM-MP can also be mounted on a CAB-MP mounting plate. Refer to Figure 10.

1. To mount the VPM-MP in a CAB enclosure or on a CAB-MP, place three of the four #10-32 x 3/8 pan Phillips screws supplied with the VPM-MP in the studs in the enclosure backbox or CAB-MP that line up with the bottom two and top left keyhole openings in the VPM-MP. Tighten the screws halfway.
2. Place the VPM-MP mounting plate over the three screws and drop it in position.
3. Install the fourth #10-32 x 3/8 pan Phillips screw into the upper right hand mounting hole. This screw will pass through the VPM mounting hole and into the stud in the CAB or CAB-MP.
4. Tighten the screws to secure the VPM-MP in place.

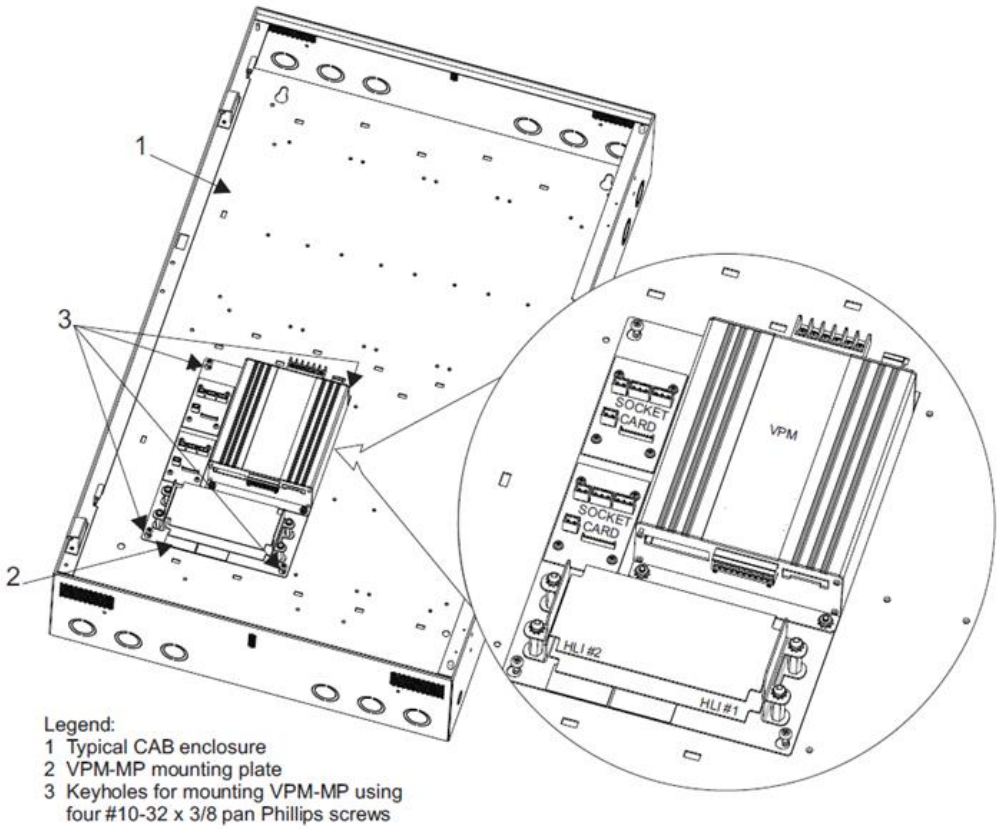


Fig. 11: Mounting the VPM-MP in a Typical CAB Enclosure

6 Electrical Ratings

	VPM	N.A.
24V Back Plane Current	N.A.	N.A.
Screw Terminal 24V Current	50mA Max. 110mA Max	110mA Max.
6.2V Back Plane Current	N.A.	N.A.
24V Standby Current	50mA Max.	110mA Max.
Output Power		
HNET/VESDAnet® Network Pair	8V peak to peak max.	8V peak to peak max.
	75mA max. (during msg transmission)	75mA max. (during msg transmission)

Wire in accordance with local code and NFPA 70. All power limited fire protective signaling conductors must be separated by a minimum of ¼" from all the following wiring located within the control panel.

1. Electric Light
2. Power
3. Class 1 or non-power limited fire protective signaling conductors

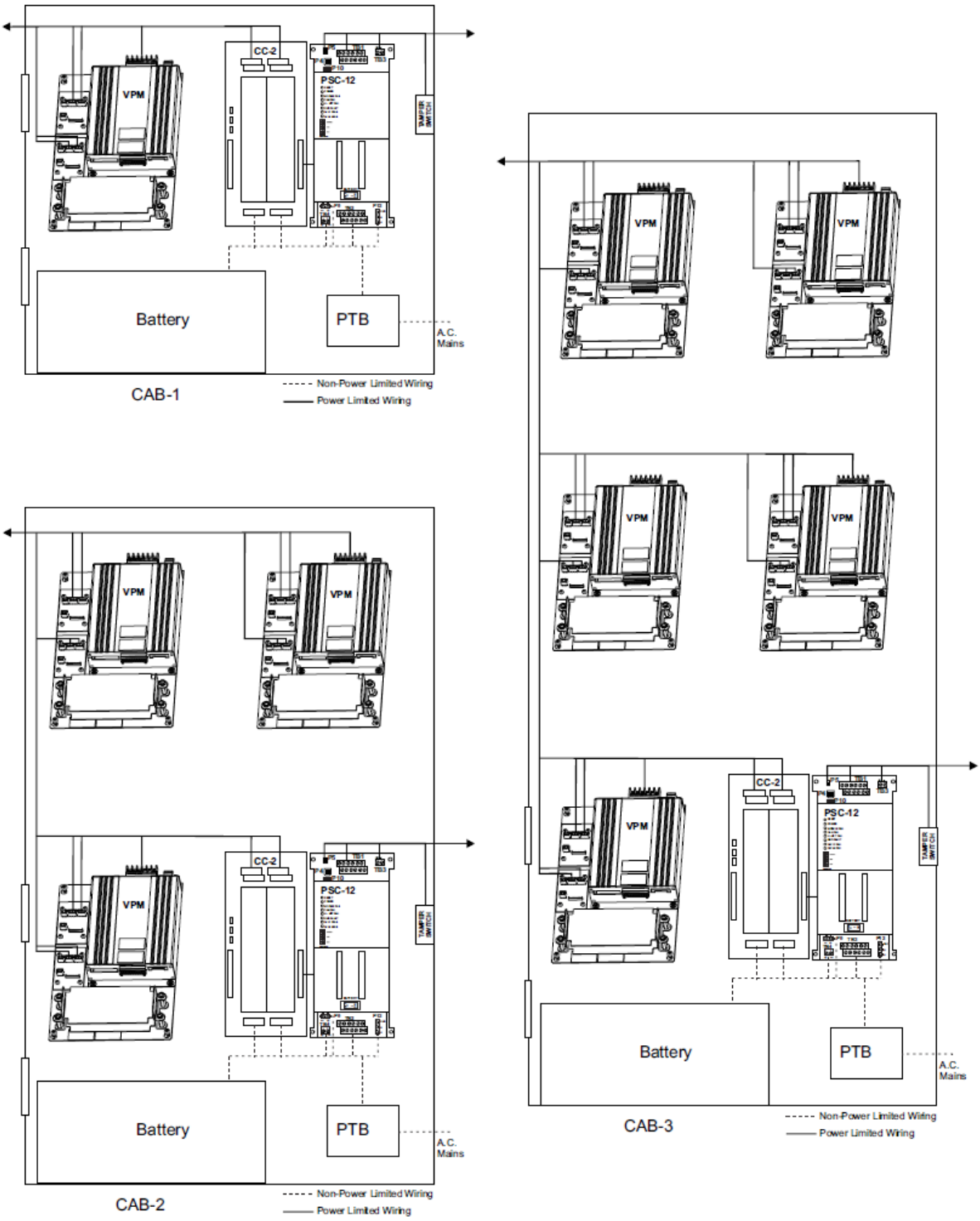


Fig. 12: Power Limited Wiring Separation

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